

WASP-52b

R-1294

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211026 · created 2026-07-31T16:34:20+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459513.7132 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.016
Transit depth (Rp/Rs)^2	2.54 +/- 0.51 [%]
Orbital Inclination (inc)	88.1 +/- 1.15
Airmass coefficient 1 (a1)	1.122 +/- 0.017
Airmass coefficient 2 (a2)	-0.097 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	#3 - [274, 180]
Optimal Aperture	3.94
Optimal Annulus	8.71
Transit Duration (day)	0.0851 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.016 vs 0.1646 (deviation 0.35σ)
Duration fit vs published	0.0851 d vs 0.0758 d (deviation 2.99σ)
Mid-time O–C	-4.5 min
Depth SNR	9.9
Residual scatter	1.54 %

Light curve

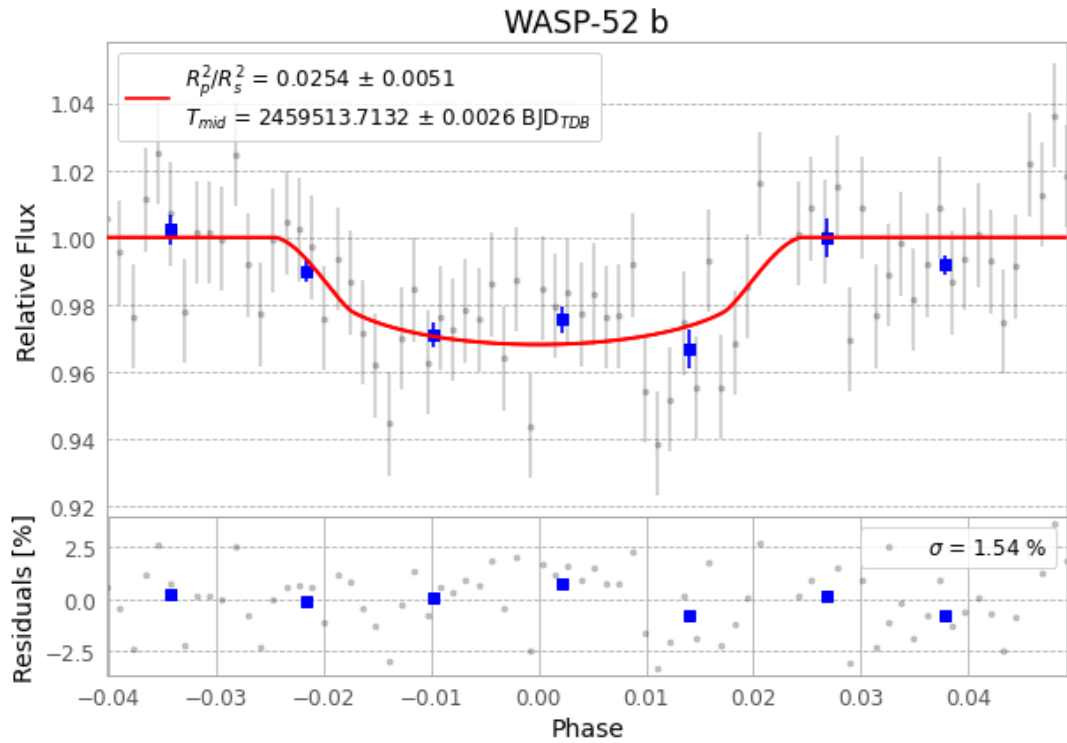


Figure 1 — FinalLightCurve_WASP-52 b_2021-10-25.png (SHA-256 1d41fd03d30d442b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [483, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 da2492fb4b0b1bc3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-52211026031812.FITS → WASP-52211026070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211026.log (ee9b6c918ae1...), FinalLightCurve_WASP-52 b_2021-10-25.png (1d41fd03d30d...), FinalLightCurve_WASP-52 b_2021-10-25.csv (a7081968e97c...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:34Z.